a different geological history than other deposits of south-west England (Dunning, F. W., personal communication). These samples will be discussed more fully elsewhere.

In contrast, groups 4 and 5 have considerably higher model ages than groups 1, 2 and 3, suggesting that the deposits owe their origin to Caledonian activity. Significant differences in the uranium-238/lead-204 and thorium-232/uranium-238 ratios are also found, implying slightly different geochemical environments.

The galenas of group 5 belong to the small, distinctive mineralization associated with Caledonian granites in the Abergairn and Ballater areas of Aberdeenshire[1]. Three galena samples of group 4 occur in the high-temperature gold–copper deposits of the Harlech Area, north Wales, which Dunham[1] was tempted to assign to the Caledonian epoch, although positive evidence was lacking. The two remaining samples come from typical Plynlimmon localities, some miles east and north-east of Aberystwyth where, significantly, the veins occur in Silurian strata. O. T. Jones[12] had already suggested that in this area some of the vein-fractures were contemporaneous with Caledonian folds.

More isotopic analyses of British and Irish common leads are in progress; detailed results and their bearing on geological problems will be published in due course.

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Palæomagnetism of Certain Rocks from Newfoundland

During the course of a visit to Newfoundland in the summer of 1957, orientated rock samples were collected by members of the Durham University Exploration Society. Pre-Cambrian rocks near St. Johns and the Mississippian Codroy rocks north of Robinson's River were sampled.

The Pre-Cambrian rocks were obtained from two localities, Signal Hill and Blackhead, on the Avalon Peninsula. The collection made on Signal Hill where the beds dip steeply eastwards represented an older horizon than the less-steeply dipping rocks collected from Blackhead, for it is clear from both graded and current bedding that the rocks are not inverted. Lithologically there is no marked difference between the two areas, the sediments at both places consisting of hard, purplish sandstones although the stratigraphical separation may be of the order of 8,500 ft.

Table 1. Mean Direction of Magnetization of Newfoundland Samples

Formation	Declination	Inclination	α	κ	N	S	Mean intensity of magnetization
Codroy group	166°	+8°	8·4	38	36	9	$16{\cdot}2 \pm 11{\cdot}4 \times 10^{-6}$
Pre-Cambrian Blackhead collection	262	+39	13·3	8	76	19	$14{\cdot}4 \pm 7{\cdot}9 \times 10^{-6}$
Signal Hill collection	232	+51	9·9	25	40	10	—
	283	+20	11·3	21	36	9	—

α, Radius of the circle of confidence; κ, a measure of precision; N, number of measurements; S, number of samples measured.

Table 2. Pole Positions calculated Before and After 20° Rotation

Formation	Site	Ancient pole	$d\psi$	$d\chi$	Re-oriented ancient pole
Codroy group	59 W. 48 N.	139 E. 43 N.	4·3	8·5	113 E. 44 N.
Pre-Cambrian Blackhead collection	53 W. 47 N.	122 W. 11 N.	9·6	16	136 W. 23 N.
Signal Hill collection	53 W. 47 N.	96 W. 5 S.	9·1	13·4	110 W. 10 N.
	53 W. 47 N.	142 W. 16 N.	6·2	11·8	163 W. 29 N.

$d\psi$ and $d\chi$, semi-axes of the oval of confidence of the pole positions.

The Mississippian rocks, some 1,200 ft. thick, comprise a lower deltaic series, the Anguille group, overlain by the Codroy group of homogeneous, fine-grained, hard red unfossiliferous sandstones. The beds dip steeply north-west, and their structural and stratigraphical relationships have been studied by Baird.

The rocks were cut and measured under an astatic magnetometer in the manner described by Collinson *et al.*[1]. The calculation of the relevant mean directions, pole positions and statistics, following the methods of Creer *et al.*[2] and Fisher[3], are given in Fig. 1*A* and in Tables 1 and 2. In these calculations the means of four measurements per sample are used and not individual measurements.

The difference between the mean directions of the two Pre-Cambrian collections is quite marked, and consequently all relevant calculations have been made for each set as well as the means for the whole group. In view of the size of sample and the uncertainty of the stratigraphic relationship of the two areas it would seem unwise to attribute much significance to this result. It is of interest, however, that this group mean pole position is fairly close to one of the Hakatai Shale results quoted by Creer *et al.*[2], and quite different from the Keweenawan of Du Bois[4].

The pole position obtained from the lower Carboniferous rocks lies considerably to the east of those quoted by Creer *et al.*[2] and Martinez and Howell[5]. This difference disappears if it could be assumed that in post-Carboniferous times there had been a relative anticlockwise rotation of Newfoundland with respect to the rest of North America. This is illustrated in Fig. 1*B*, where a rotation of 20° has been assumed.

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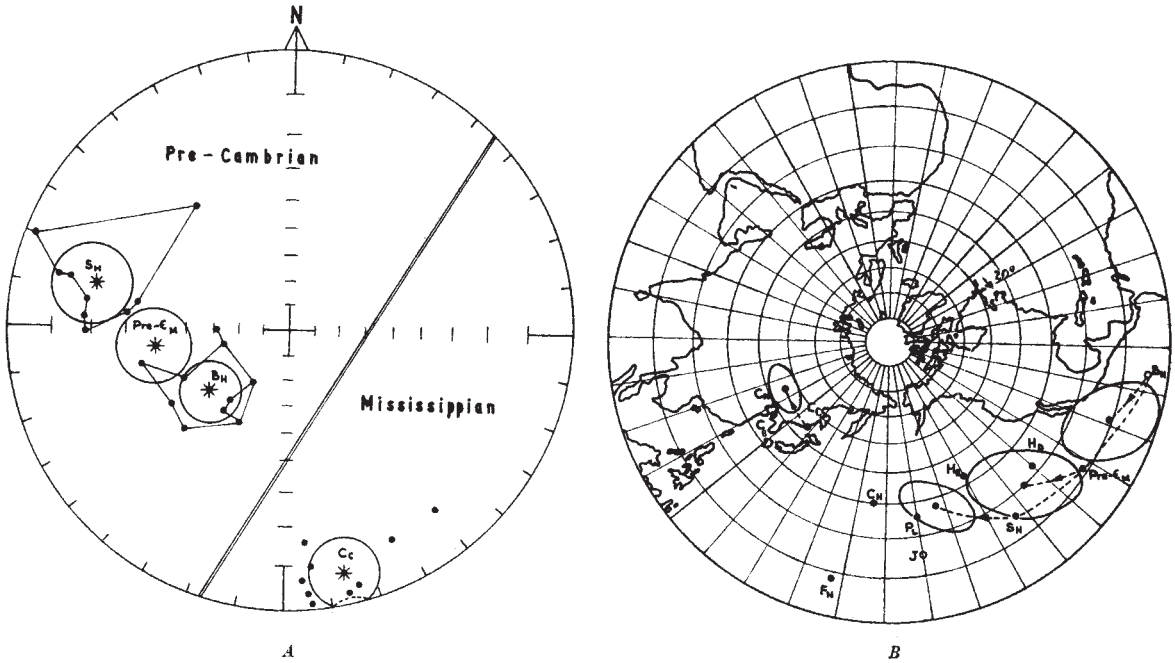


Fig. 1*A*. Stereographic plot of the directions of magnetization of Pre-Cambrian and Carboniferous (Mississippian) samples from Newfoundland. (Signal Hill and Blackhead samples separately linked.) All north-seeking poles point down. * represents the mean direction enclosed within the circle of confidence

Fig. 1*B*. Plot of pole positions of American Pre-Cambrian and Carboniferous rocks, illustrating the effect of a 20° rotation on the Newfoundland samples. Ellipses of confidence are drawn around the Newfoundland pole positions. Pre-ε_M mean of all Newfoundland Pre-Cambrian (S_H, Signal Hill; B_H, Blackhead collections). H_{D_1} and H_R pole positions of Hakatai Shale samples according to Doell and Creer *et al.* P_L, C_H, FN, J, Portage Lake, Copper Harbour, Freda and Nonesuch, and Jacobsville divisions of the Middle and Upper Keweenawan. (Jacobsville and Blackhead poles are south of the equator.) C_C pole position of Codroy samples; C_N and C_B pole positions of Naco Sandstone and Barnet Shale samples

The effect of such a rotation upon the Pre-Cambrian pole positions is to improve greatly the agreement with the rest of North America. The group mean position approaches those obtained by Runcorn and Doell for the Hakatai Shales, while the new pole position of the Signal Hill samples even suggests possible correlation with the Portage Lake horizon of the Keweenawan[4]. This is clearly seen in Fig. 1*B*, where the re-oriented pole positions are enclosed within their ellipses of confidence.

This suggested rotation is based entirely on palæomagnetic considerations; whether or not it is entirely fortuitous depends upon geological considerations, and it appears (Westoll, T. S., personal communication) that such a movement has been suggested by some geologists.

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PHYSICAL SCIENCES

Anomalous Night-Time Reception of a Major Solar Radio Burst

During the early morning hours of March 8, 1958, the planet Jupiter was under observation by a total of five different receiving arrays at the University of Florida Radio Observatory. Two of these arrays operated at a frequency of 18 Mc./s., two at 22·2 Mc./s., and one at 27·6 Mc./s. Both pen recording and aural monitoring of the signals by an observer were employed. At 0235 U.T. an unusual event began simultaneously on all five channels, consisting of a single very intense burst of noise which slowly rose to a maximum, and even more slowly declined over a total period of about 2 min. During this interval the observer was able to obtain an excellent record of the polarization of the disturbance with the 22·2 Mc./s. polarimeter, which was kept in stand-by condition during all periods of observation.

The outburst differed from typical Jupiter noise in a number of respects: (1) the relatively smooth rise and fall of the signal was slower by at least an order of magnitude than any previously observed single Jupiter pulse[1]; (2) no Jupiter pulse had been observed at this Laboratory simultaneously on all three frequencies; (3) analysis of the polarimeter record showed the burst to be polarized in the left-hand elliptical sense, whereas all the Jupiter noise analysed here had shown right-hand circular or elliptical polarization. Although the event had occurred more than four hours prior to sunrise, it was

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